

A formal knowledge base for metagenomics and geochemistry of the Rub' al Khali desert

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Abstract

Integrating multi-omic data with environmental metadata remains a significant challenge due to the heterogeneity of data formats and a lack of standardized semantic representations. Building such an integrated resource requires addressing several interlocking challenges: defining a unified data model across instruments and observation types, harmonizing taxonomic identifiers across reference databases, validating incoming data against shared schemas, and supporting semantic query answering at scale under a tractable reasoning regime. Here, we present a high-resolution microbiome dataset from five expeditions across a 1,043 km transect of the Empty Quarter (Rub' al Khali) of Saudi Arabia, comprising 2,516 sample individuals from 60 primary sampling sites and special-interest locations. A 2,550-row source ledger connects specimens, DNA extractions, libraries, sequencing runs and community profiles; the canonical feature table contains 1,271 taxonomic profiles, of which 1,237 pass ecological filtering. These biological sequences are joined to 274 field environmental logs, 12,936 monthly climate measurements, 712 archived-soil pH observations, and two X-ray fluorescence workflows (71 field sessions and 725 laboratory records). We augment these data with a formal knowledge base constructed using OWL 2 DL and the Semanticscience Integrated Ontology (SIO), with ShEx-based ingest validation and ELK-based consistency checking over the resulting graph. This semantic layer enables complex cross-domain queries and provides a comparative baseline for studying microbial adaptation in extreme environments.

Keywords: Knowledge Graph, Metagenomics, Rub' al Khali, SIO, Semantic Web

1 Background and summary

The Rub' al-Khali (Empty Quarter) is the largest contiguous sand desert on Earth, a hyper-arid sand sea of approximately 500,000 km² across the southern Arabian Peninsula [1]. Air temperatures recorded during our expeditions spanned 20.0–49.4°C across 273 field observations, and the region receives negligible and highly irregular rainfall. Despite these extremes, this environment harbors microbial communities whose composition, diversity, and ecological drivers remain poorly characterized [2]. To predict ecosystem responses to climate change and to inform conservation and land-use policy, researchers require deep knowledge of the biogeochemistry of such extreme environments [3].

Saudi Arabia already has a substantial arid-soil microbiome literature. Previous studies examined elevation gradients [4], rainfall responses across soil compartments [5], open and rhizosphere soils across three regions [6], rhizosphere bacteria of pioneer sand plants near Jizan [7], and bacteria associated with native desert plants [8]. More recent surveys characterized fertility islands and plant-associated networks [9, 10], sandy soils near natural and artificial water points [11], rhizosphere and endosphere communities across northern reserves [12], and Red Sea coastal sabkhas [13]. Fungal studies progressed from a small number of western-coastal and open-sand samples [14, 15] to a standardized single-visit survey of 85 locations across Saudi Arabia

[16], and a recent synthesis identifies substantial geographic and taxonomic gaps in Arabian biological-soil-crust research [17]. In the United Arab Emirates, related studies profiled a Rub’ al-Khali interdune microbial mat [18] and desert-plant rhizosphere and endosphere communities [19]. These studies establish that a Saudi arid-soil survey is not itself new. What remains missing is a repeated, multi-expedition transect in which every community profile stays joined to its provenance across locations, physical specimens, molecular products, environmental observations, climate, and geochemistry.

To characterize complex ecosystems, researchers must integrate heterogeneous and multi-scale data: microbial community profiles derived from DNA sequencing, elemental soil composition from geochemical analyses, spatiotemporal environmental variables, and provenance metadata that link physical samples to their digital representations. Traditional approaches store these data types in separate, disconnected repositories, which limits cross-domain queries and reduces reproducibility.

Engineers originally developed the concept of a digital twin, and environmental scientists increasingly adopt this approach [20, 21]. Recent frameworks, such as TwinEco, standardize the development of dynamic, data-driven digital twins in ecology to avoid fragmented software designs [22]. Additionally, regional natural resource management benefits from artificial intelligence-assisted digital twin architectures that unify heterogeneous environmental observations [23]. Such digital twins show strong predictive accuracy when they incorporate continuous data streams, such as citizen science observations for real-time biodiversity forecasts [24]. To guarantee that these models remain interoperable and reusable across domains, researchers emphasize the integration of datasets, models, and computational workflows through Findable, Accessible, Interoperable, and Reusable (FAIR) principles [25]. However, while biological systems and specific regions have digital representations [26], researchers have not yet developed semantic representations of entire desert ecosystems. Furthermore, ecosystem digital twins largely focus on macroscopic scale and do not go across levels of granularity.

Constructing such a resource for an entire desert ecosystem requires addressing several interlocking challenges that have shaped the design of this work. First, the data are produced by heterogeneous instruments and protocols (16S amplicon sequencing, X-ray fluorescence, GPS-tagged field notes, gridded climate reanalysis), so a *unified data model* is needed that can represent samples, processes, qualities, and quantities under a common upper-level vocabulary rather than as disconnected tables. Second, taxonomic labels are reported by multiple primer choices and classifier workflows and must be *harmonized* against a single reference taxonomy without losing the original assertions. Third, because data are deposited in batches over five field expeditions, *schema validation at ingest* is required to catch structural errors before they propagate into the graph. Fourth, the knowledge graph must support *semantic query answering at scale*: with tens of millions of triples, expressive OWL 2 DL reasoning is computationally infeasible, so a tractable reasoning regime must be selected and its trade-offs made explicit. Fifth, a *user-facing query and exploration interface* is required so that domain scientists who are not fluent in SPARQL can interact with the resource. We return to each of these challenges in the usage notes (Section 6) and link them to the corresponding design choices in the methods and knowledge representation sections.

Community reporting standards and knowledge-graph infrastructure define the reuse target this resource is built against. The FAIR principles set the criteria a deposited resource is judged against [27]. MIXS/MIMARKS define contextual fields for marker-gene and environmental sequence data [28], and the STREAMS guideline extends reporting practice to environmental microbiomes [29]. The National Microbiome Data Collaborative shows how persistent identifiers, standardized metadata and linked-data schemas support reuse across environmental microbiome studies [30]. On the knowledge-graph side, KG-Microbe connects microbial taxa with traits, environments and growth preferences [31], and MetagenomicKG integrates taxonomic, functional and biomedical sources for metagenomic applications [32]. These resources establish that a microbiome knowledge graph is not itself novel. The contribution of the present resource is the joined, expedition-level provenance of locations, physical specimens, laboratory processes, geochemistry, climate, sequence runs, quality control and taxon observations from a single landscape survey, rather than vocabulary coverage or graph construction alone.

Here, we develop a knowledge graph of large parts of the Rub’ al-Khali desert ecosystem to serve as the semantic foundation of the Rub’ al-Khali digital twin. We present an OWL 2 DL knowledge graph [33] built from RDF instance data [34]. This knowledge graph integrates data from five field expeditions (2023–2025) along a transect of more than 1,000 km across the Saudi Rub’ al-Khali, from the western escarpment to the eastern border with Oman. A versioned source ledger of 2,550 rows (2,302 master-sheet rows and 248 plant rows) links field specimens to their downstream products; 2,516 non-control rows are represented as sample individuals in the graph. The canonical 16S rRNA gene amplicon feature table contains 1,271 profiles, of which 1,237 pass ecological filtering across 64 numeric site labels. We incorporated X-ray fluorescence (XRF) measurements from two complementary workflows: 71 complete in-situ handheld field sessions at 58 Trip 5 sites, and 725 processed laboratory records of archived material from all five expeditions. We further integrated site-specific environmental metadata recorded at each visit (274 field logs), 712 admitted archived-soil pH observations, and monthly climate records derived from the Open-Meteo database [35].

We structured the knowledge graph with the Semanticscience Integrated Ontology (SIO) [36] as the upper-level framework. This design allows interoperability with other SIO-based datasets. We modeled domain-specific concepts as subclasses of SIO entities, and all measurements follow SIO’s ontology design pattern for measurement processes. We grounded chemical entities in the Chemical Entities of Biological Interest (ChEBI) ontology [37], environmental contexts in the Environment Ontology (ENVO) [38], and taxonomic identifiers in the NCBI Taxonomy [39].

Because it encodes the full provenance chain from physical sample collection to digital data products, the knowledge graph is the key foundation for implementing the FAIR data principles for the Rub’ al-Khali digital twin. This semantic integration eliminates the data fragmentation that typically limits ecological models, and it guarantees that observations, models, and workflows remain fully interoperable [25]. Consequently, the knowledge graph acts as the operational foundation for the Rub’

al-Khali digital twin. It allows researchers to execute cross-domain queries that otherwise demand complex joins across multiple heterogeneous tables in a conventional database. Therefore, this resource serves as a foundational infrastructure for computational ecology, which facilitates hypothesis generation, intervention simulation, and evidence-based policy.

This resource makes several contributions. On the scientific side, it provides a coordinated multi-trip desert microbiome and geochemistry dataset—16S rRNA amplicon profiles, XRF elemental measurements, archived-soil pH, GPS-tagged field metadata, and monthly climate records—for 2,516 sample individuals from five expeditions across a 1,043 km transect of the Rub’ al-Khali, together with a harmonized taxonomic abundance table that maps amplicon sequence variants onto NCBI Taxonomy while preserving the original assertions. On the technical side, the resource comprises an OWL 2 DL knowledge graph built on SIO with domain classes grounded in ChEBI, ENVO, and NCBI Taxonomy, modelling the full provenance chain from field sampling to digital data products. We formalize reusable ontology design patterns for sampling, measurement, taxonomic abundance, and provenance aligned with SIO’s core idioms. A ShEx-based ingest validation workflow enforces structural shapes on every RDF batch before loading, and the ELK reasoner is used to validate logical consistency of the knowledge graph. A set of competency questions documents the intended query workload and serves as regression tests, and a web portal with an interactive map, per-site dashboards, and a SPARQL editor exposes the knowledge graph to domain scientists without requiring SPARQL fluency. The companion ecological analysis reports the inferential results for the quality-controlled community profiles, laboratory-XRF gradient, rainfall, spatial turnover and association networks [40]; the present paper describes the reusable data products, provenance model, identifier mappings and their executable validation.

2 Methods

2.1 Study area and sampling design

We conducted five sampling expeditions across the Rub’ al-Khali along a transect from the western escarpment (19.15°N, 45.17°E) to the eastern border of Saudi Arabia with Oman (20.71°N, 54.73°E). The 60 ordered sampling sites yield a 1,043-km cumulative path and lie 1,015 km apart end to end. Figure 1 illustrates the altitude profile across this transect.

The planned repeated transect encompasses 60 primary sampling sites at predetermined geographic coordinates, alongside 10 named sites of special interest (water wells, a hot spring, an oil spill site). Four additional numeric identifiers (61–64) were used only on Trip 1; coordinate matching against the field records establishes that these denote four of the named catalogue locations rather than additional physical sites. We retain the numeric identifiers as source provenance but do not count them as four additional locations, and we exclude their records from inference over the intended repeated-campaign frame of sites 1–60 because those regions could not be revisited.

To capture temporal variability, we conducted expeditions across different seasons: late winter 2023 (Trip 1, 64 sites visited), summer 2023 (Trip 2, truncated to 8 sites due

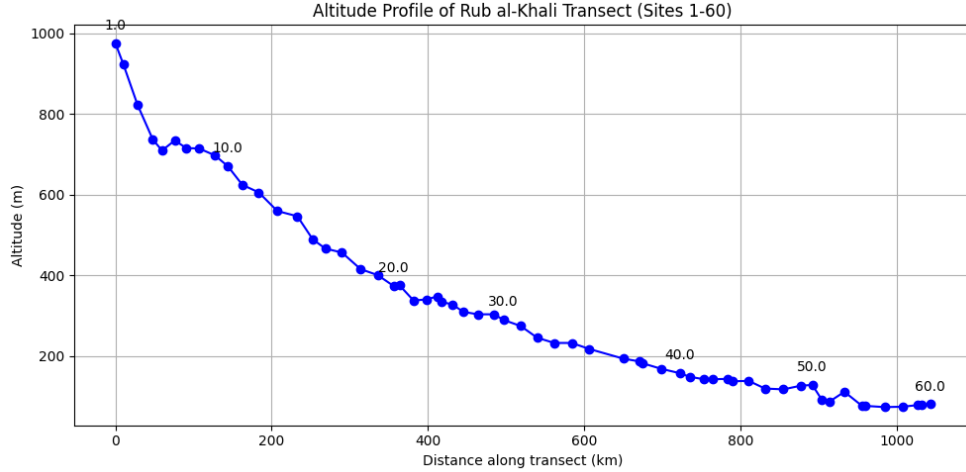


Fig. 1 Altitude profile along the sampling transect across the Rub’ al-Khali, from the western escarpment to the eastern border with Oman.

to extreme temperatures), winter 2024 (Trip 3, 65 sites visited), summer 2024 (Trip 4, 60 sites visited), and autumn 2025 (Trip 5, 62 sites visited). We summarize the observational coverage in Table 1. We recorded temperature, atmospheric pressure, and relative humidity at each visited site, although the available measurements differ by expedition. We acquired X-ray fluorescence (XRF) data through two separate workflows that are not interchangeable replicates: handheld in-situ measurements during Trip 5, and laboratory measurements of archived material from all five expeditions.

Table 1 Coverage of the field environmental logs. Named and auxiliary records are not added to the numeric site catalogue as if they were necessarily distinct physical locations.

Expedition	Season	Field-log records	Measurements available
Trip 1	Late Winter 2023	1–60; 4 named; 15 auxiliary*	Temp, Press, Hum.
Trip 2	Summer 2023	1–8	Temp.
Trip 3	Winter 2024	1–60; 5 named	Temp, Press, Hum.
Trip 4	Summer 2024	1–60	Temp, Hum.
Trip 5	Autumn 2025	1–60; 2 named	Temp, Press, Hum.

*The 15 records are appended to the legacy Trip 3 worksheet but are dated March 2023 in the original workbook and are curated as Trip 1 auxiliary/revisit observations.

The curated environmental table is generated from the immutable field sheets through an exact-cell correction ledger and retains all 274 nonblank records. Where the source is unambiguous, the ledger restores it: the Trip 2 sheet records eight values in the humidity column while the temperature column is empty, and the field note that the trip ended because of extreme temperature supports reading them as temperatures. The Trip 5 site 40 source humidity of 194% is outside the physical range and remains

missing until a primary source establishes a correction. The graph generator links 260 of the 274 records to catalogue site visits, yielding 260 temperature, 187 pressure and 251 humidity values; the remaining 14 auxiliary or revisit labels denote offset or special-purpose locations and remain table-only, because coincident coordinates are not treated as evidence that they are identical to a catalogue site.

At each site, we collected samples from three distinct soil compartments: bulk surface soil (0 cm depth), bulk subsurface soil (5–10 cm depth, hereafter “deep”), and subsurface soil collected near roots (5–10 cm depth, hereafter “root-adjacent”). The legacy label “deep” and the class label “Deep Soil Sample” denote the 5–10 cm bulk compartment and do not imply a greater sampling depth. We obtained three co-located field replicates per compartment by placing soil into sterile tubes, either by scooping soil directly into the tube or using sterile plastic spoons; these replicates are not independent sites. We preserved samples at -20°C in the field using an Engel MR040F portable freezer. At the end of each expedition, we transferred samples to an -80°C freezer in the laboratory.

We included positive and negative controls for each sampling campaign to mitigate and control for contamination artifacts. The positive-control design differed by expedition. Trips 1 and 2 used the ZymoBIOMICS HMW DNA Standard D6322, supplied as purified high-molecular-weight DNA and therefore entering after extraction. Trip 3 used the whole-cell ZymoBIOMICS Microbial Community Standard D6300 and carried it through extraction. Trip 4 had no positive control. Trip 5 used D6300, but its positive and negative control pair was sequenced by shotgun metagenomics rather than by the 16S assay. D6322 contains seven bacterial genomes at 14 % genomic DNA each and *Saccharomyces cerevisiae* at 2 %; D6300 contains eight bacterial genomes at 12 % each and two yeasts at 2 % each. These manufacturer genomic-DNA fractions are composition specifications, not expected 16S read fractions.

Negative controls included distinct extraction blanks and PCR blanks. One extraction blank was included per extraction day, and an extraction day could contain samples from several expeditions, so an extraction blank is linked to its extraction batch and never directly to a trip. The Trip 4 workbook records five numbered blanks and one unnumbered blank, but the blank covering a three-day extraction was not sequenced, leaving all or part of 23 sites without a sequenced blank. Author-confirmed control facts are frozen in a ground-truth table; identity, batch-membership and assay-applicability claims that await laboratory confirmation are represented in the graph as evidence-bearing dispositions rather than as ground truth, and no field-blank instance is asserted because no sterile-bag inventory was recovered.

2.2 DNA extraction and sequencing

We utilized 0.9–1.2 g of soil for DNA extraction using the PowerLyzer PowerSoil Kit (QIAGEN, catalog #12855-50) or the DNeasy PowerSoil Pro Kit (QIAGEN, catalog #47014). We homogenized all samples with the PowerLyzer 24 Homogenizer and completed extraction either manually or via the automated QIAcube Connect system. We quantified the eluted DNA using the Qubit HS dsDNA kit.

We amplified bacterial DNA using the Bakt_341F (CCTACGGGNGGCWGCAG) and Bakt_785R (GACTACHVGGGTATCTAATCC) primer pair [41] to target the V3–V4 region of

the 16S rRNA gene. Both primers carried Illumina overhang adapter sequences. We performed amplification using the Multiplex PCR kit (QIAGEN, catalog 206143) with an initial activation at 95 °C for 15 min, 35 cycles of 94 °C for 30 s, 57 °C for 90 s and 72 °C for 90 s, and a final extension at 72 °C for 10 min, and sequenced the resulting pools on an Illumina NovaSeq 6000 platform. We processed all samples in the PI’s laboratory at KAUST.

The reverse primer identity is a correction to the previously recorded protocol, which named the Apprill 806RB primer (GGACTACNVGGGTWTCTAAT). A bounded audit of the raw reads resolves the discrepancy against the recorded value: streaming the first 10,000 reads of three Trips 1–4 and three Trip 5 biological read pairs, 56,891 of 60,000 forward reads (94.82%) begin with Bakt_341F and 55,057 of 60,000 reverse reads (91.76%) begin with Bakt_785R, whereas 11 reverse reads (0.018%) match the 806RB sequence. The staged protocol record carries the corrected pair together with the superseded entry, so the change is auditable rather than silent.

2.3 Sequencing data processing

We processed raw amplicon reads employing nf-core/ampliseq (v2.14.0) [42, 43], applying the DADA2 algorithm for error correction and denoising and assigning taxonomy against the SILVA 138.2 reference database [44]. A portable samplesheet regenerated from the deposited submission records covers 1,268 submitted run rows over 1,238 distinct sample identifiers, of which 30 carry a second library.

The canonical Trips 1–5 feature table is the direct output of that pipeline. It is unfiltered: it retains all 351,472 amplicon sequence variants (ASVs) and still contains the 24 control profiles, and no contaminant removal has been applied to it. A QIIME2 [45] *decontam* branch exists in the analysis tree, but it operated on a different and smaller table, and in the form that survives the command that would have produced the contaminant table is commented out while a later step consumes that missing artifact. No removal fraction from that branch is defensible, and we do not restate a prevalence threshold as an applied filter.

We instead generated an assay-aware audit following low-biomass control recommendations [46]. The primary test pools the 17 extraction-day blanks with frozen links to Trip 5 profiles and uses an uncorrected one-sided Fisher $p < 0.1$, requiring presence in at least two blanks; because one blank represents each extraction batch, within-batch prevalence estimation is impossible. The resulting 351-ASV candidate set is applied only to the 217 canonical biological profiles linked to those batches. Seven unambiguously positive-labelled 16S profiles from Trips 1–3 are used to evaluate expected bacterial-genus recovery, never to train the contaminant classifier. The primary table remains unfiltered, and the candidate set is used only in a bounded sensitivity analysis (Section 6).

Genome-encoded functional potential used for the companion analysis was predicted with PICRUSt2 v2.4.1; its execution manifest, with the exact commands, hidden-state method, edge exponent, seed and thresholds, is archived with the release.

All deterministic metadata corrections are recorded without overwriting the source values. For example, in three records the value NEBNext had shifted into the site column while the other populated cells remained aligned; the confirmed correction

ledger restores site 46, records **NEBNext** as the displaced library-preparation kit value and retains the original cell value and rationale.

2.4 Taxonomy normalization and mapping

Amplicon classifications were assigned against SILVA 138.2. We retain the original classifier lineage and ASV identifier as the primary evidence. The combined Trips 1–5 export contains two source encodings: the seven SILVA rank fields alone, or those seven fields followed by a separate species-assignment field and a numeric confidence. QIIME artifact provenance identifies the latter as an appended species assignment rather than an eighth rank. We therefore normalize only the ordered domain–species fields: rank prefixes are removed, empty and unclassified values are represented explicitly, and the separate species assignment is retained in a provenance ledger but never promoted into the asserted lineage.

A fail-closed normalization step evaluates the historical SILVA-to-NCBI candidates against a pinned NCBI Taxonomy release (the OBO Foundry NCBITaxon release 2025-03-13, built from the NCBI organismal taxonomy of 2025-03-01). An NCBI identifier is retained only when it resolves, has a compatible rank and label or documented synonym, and its ancestor chain is compatible with all retained higher-rank identifiers in that source lineage. A missing, ambiguous or incompatible candidate is replaced by a deterministic project class scoped to the complete truncated source lineage and rank, rather than by the nearest ancestor’s identifier, which would conflate a named child with its broader ancestor. Each decision records the source lineage and rank, candidate identifier, validation outcome, final identifier and reason. The output mapping must cover every canonical lineage represented in both the Trips 1–5 taxonomy export and the feature table; otherwise generation stops.

Existing project identifiers are retained only when all explicit historical uses share one normalized rank, label and nearest explicit parent context; a reused identifier that acquires more than one corrected context is replaced in every affected use by contextual classes. Conflicting historical evidence is therefore contextualized rather than resolved according to file order. The mapping builder emits corrected mappings, a decision ledger, a clean project taxonomy module and a manifest containing input and output checksums, and the abundance-ABox generator accepts only a passing, checksum-matched manifest.

We evaluated cross-resource alignment to GTDB [47] and iNaturalist separately from this SILVA-to-NCBI layer, in an earlier exploration whose candidate tables are not part of the release. The conclusion that exploration supports is a negative one: homonyms, *Candidatus* names, placeholder taxa, GTDB split taxa, rank conflicts and database-version drift can all produce identical labels for non-identical concepts, so label agreement alone does not establish extensional identity. The canonical release modules therefore contain no `owl:equivalentClass` assertions generated from those lexical candidates. Unreviewed GTDB and iNaturalist candidates are excluded from the asserted crosswalk, and we claim no mapping precision or recall because a stratified manual gold standard has not been completed. This conservative representation

permits queries over the original SILVA lineage and validated NCBI classes without turning lexical similarity, a shared label or a supplementary species call into unsupported taxonomic identity.

2.5 Environmental and climate data

We retrieved historical climate data from the Open-Meteo archive endpoint [35] for each recorded field location, yielding 12,936 monthly records encompassing temperature, total precipitation, rainfall and relative humidity, together with annual source fields. Retrieval keyed on the **Site** column of the per-trip geodata sheets, a field-notebook column rather than an integer primary key. Of the 81 distinct fetch keys, 66 resolve to a site individual (all 60 numbered sites and six named ones), which gives the released denominator of $66 \times 49 \text{ months} \times \text{four variables} = 12,936$ monthly records. The remaining 15 keys carry an unrecorded offset or free-text label (for example **47+grass**, which records a reading taken near site 47 rather than a reading of it) and are quarantined rather than mapped onto their numeric stem. Exact site-key resolution retains 582 annual values in the graph and quarantines 54 values attached to unresolved field-notebook labels. Furthermore, field researchers manually classified each site according to its biome (e.g., desert biome) and specific environmental features (e.g., sand dune, saline pan, gravel, desert oasis). We subsequently mapped these classifications to the Environment Ontology (ENVO) [38].

2.6 XRF acquisition, processing and reconciliation

We acquired XRF data through two complementary workflows. During Trip 5, researchers used a Vanta handheld instrument to perform in-situ field measurements. The source log contains 106 entries across 59 sites; 71 complete sessions at 58 sites were retained, and nine of those sites have repeated complete sessions. The field log carries no physical-specimen, compartment or depth identifier, so field and laboratory linkage is site-level rather than a same-aliquot comparison. Archived material from all five expeditions was subsequently measured in the laboratory. The canonical laboratory inputs contain 547 selected sample records from Trips 1–4 and 178 from Trip 5 (725 records in total), and all 725 enter the analyses. A former 158-row Trip 5 table and the resulting 705-row cross-trip table are retired legacy artifacts and are not analytical inputs.

The provenance audit reproduced all 13,390 reported Trips 1–4 cells under the current largest-positive-value rule and all 4,293 reported Trip 5 cells under the parser’s different last-reported-value rule. Neither rule is treated as instrument-prescribed, and the release discloses the two source-specific policies rather than implying a common calibrated procedure. All instrument channels were mapped against a pinned ChEBI release and a dated PubChem snapshot; an executable audit rejects missing, deprecated or formula-mismatched records, and unsupported mappings remain null. For a descriptive cross-workflow diagnostic, repeated field sessions were reduced by their site median and matched to the Trip 5 laboratory record at 58 sites; rank agreement was generally modest. No cross-workflow calibration is inferred, and no analysis assumes that field and laboratory values have the same scale.

2.7 Archived-soil pH acquisition and admission

We measured archived dry soil after sieving below 2 mm. Each assay used an independent 1 g scoop suspended in 2.5 mL 0.01 M CaCl_2 , with a two-point pH 7/10 calibration, electrode slope and pH-10 buffer read-back recorded per session; admission required calibration marked yes, slope 95–102 % and a read-back within 0.1 pH units of nominal. All measurements available when laboratory follow-up ended were frozen as the versioned dataset `EQ-PH-SHARED-v1.0.0`, the sole pH version used by this descriptor and the companion ecology paper.

The audit accounts for all 1,168 source rows: it admits 712 measurements, retains 356 rows without a measurement, marks 45 as depleted, quarantines 36 with ambiguous dates, and excludes 19 numeric rows with incomplete recorded quality control. We neither inferred missing quality-control fields nor imputed unavailable values. The workbook has no session identifier, so we reconstructed 29 session individuals from unique combinations of trip, accepted measurement date, electrode slope and pH-10 read-back; these groupings preserve the available quality-control context but are not directly recorded laboratory batch identifiers.

2.8 Schema validation

To ensure structural integrity within the RDF graph, we used Shape Expressions (ShEx) [48]. ShEx provides a formal structural schema language that validates the presence, cardinality, and datatype of RDF properties. We defined specific ShEx shapes for all core entity types in the knowledge graph. These entity types include sampling sites, soil samples, DNA extracts, sequencing datasets, measurements, quality control records, laboratory control materials, control roles, laboratory batches, control sequence occurrences, archived-soil pH processes, values and sessions, and taxonomy mappings.

We integrated this schema validation into our deployment workflow using a Shape Map. The Shape Map functions as a binding directive, explicitly linking node selectors (e.g., all instances of the sampling site class) to their respective ShEx shape definitions. The workflow tests each newly generated segment of instance data against the schema. It terminates the deployment if it detects missing mandatory properties (e.g., GPS coordinates for a sampling site) or invalid datatypes. This constraint application guarantees that all generated data conforms to the expected structural properties. We implemented this validation using the Apache Jena ShEx library (version 4.10.0) invoked from custom Groovy scripts. The multi-gigabyte taxonomy ABox is excluded from this memory-bounded ShEx run; the exact generated file is instead parsed in full with an independent Turtle parser and checked by a streaming validator over every classification process–dataset–quality–value relationship.

2.9 Logical consistency (OWL reasoning)

We checked logical consistency using the ELK reasoner [49]. The ELK reasoner operates on the OWL 2 EL profile and detects unsatisfiable classes and contradictory axioms within ontologies. ELK implements a polynomial-time algorithm, facilitating the consistency validation of large knowledge bases. However, this performance benefit

necessitates a tradeoff in logical expressivity, as the OWL 2 EL profile does not support complex constraints such as universal quantification or inverse object properties found in ontologies like the SemanticScience Integrated Ontology (SIO). We employed ELK instead of more expressive reasoners (e.g., HermiT [50]) because verifying the large knowledge graph we created with a more expressive reasoner proved impossible due to computational complexity. The current run is memory-bounded and excludes the multi-gigabyte taxonomy ABox and large reference modules, so a successful run establishes bounded consistency of the included modules rather than whole-graph OWL 2 DL consistency. This approach enabled rapid continuous integration during ontology development.

2.10 Reproducible workflow

Release generation, validation and manuscript-derived statistics are orchestrated from one Nextflow entry point (`main.nf`), with separate local and cluster profiles that change execution infrastructure without changing process definitions. The workflow regenerates the site, environmental-measurement, sample, XRF, DNA, sequence-record, quality-control, laboratory-control and pH modules from their declared sources and requires every byte stream to match the staged release candidate, so the deposited modules and the manuscript statistics are produced from the same auditable run.

3 Knowledge representation

We developed a formal knowledge base to integrate the different types of data generated in this project. The semantic layer encodes the full provenance chain from physical sampling to digital data products, enabling cross-domain queries and ensuring data interoperability through shared vocabularies.

3.1 Architecture

The knowledge base is constructed in Description Logic [51] using a modular architecture that strictly separates the terminology (Terminological Box, TBox, or ontology) from the observed facts (Assertion Box, ABox, or domain knowledge). The TBox defines the ontology of the knowledge graph using OWL 2 DL [33], while the ABox contains the instance data generated from the expedition. We deployed the complete knowledge base across three named RDF graphs: an ontology graph (17,025 triples) containing the TBox, a knowledge base graph containing the asserted instance data, and a taxonomy reference graph containing the aligned taxonomy module.

3.2 Prefixes used in listings

For brevity, the Turtle and SPARQL listings throughout the paper use the prefix declarations of Table 2; these declarations should be prepended to any listing that the reader wishes to execute against the public endpoint. With these declarations in place, identifiers such as `rak:P100001` are abbreviated to `rak:P100001` and `sio:SIO_000291` to `sio:000291` without loss of information.

Table 2 Prefix declarations used in all Turtle and SPARQL listings.

Prefix	Namespace
rak:	https://rubalkhali.science/kb/RAK_
sio:	http://semanticscience.org/resource/SIO_
pato:	http://purl.obolibrary.org/obo/PATO_
uo:	http://purl.obolibrary.org/obo/UO_
envo:	http://purl.obolibrary.org/obo/ENVO_
ncbitaxon:	http://purl.obolibrary.org/obo/NCBITaxon_
obo:	http://purl.obolibrary.org/obo/
rdf:	http://www.w3.org/1999/02/22-rdf-syntax-ns#
rdfs:	http://www.w3.org/2000/01/rdf-schema#
xsd:	http://www.w3.org/2001/XMLSchema#
geo:	http://www.opengis.net/ont/geosparql#
dcterms:	http://purl.org/dc/terms/

3.3 Ontology engineering and design patterns

We use the Semanticscience Integrated Ontology (SIO) [36] as the upper-level framework. SIO provides general-purpose classes and relations applicable across scientific domains, ensuring semantic interoperability with other SIO-based datasets. We defined all domain-specific classes as subclasses of SIO classes, and all introduced relations are sub-relations of SIO relations. We mapped classes to SIO and maintained full semantic mappings in our public repository (see Section 8).

The canonical terminology module declares 333 classes: 297 project-local classes and 36 referenced classes (32 SIO and four PATO). It also declares 20 object properties (11 project-local and nine SIO) and 35 datatype properties (34 project-local and one SIO). These are declaration counts from the staged canonical module, not counts of the imported ontologies or of whole-graph axioms. This compact semantic structure maintains precise alignment with our standardized schemas while minimizing redundant inferences.

3.3.1 Measurement pattern

We model a quantitative measurement using SIO’s ontology design pattern for measurement processes: a measurement is a process that generates a value for a specific quality of a target entity.

$$\text{MeasurementProcess} \sqsubseteq \text{Process} \quad (1)$$

$$\text{Quality} \sqsubseteq \text{Attribute} \quad (2)$$

$$\text{Quantity} \sqsubseteq \text{InformationContentEntity} \quad (3)$$

$$\text{MeasurementProcess} \sqsubseteq \exists \text{hasTarget.Entity} \sqcap \exists \text{hasOutput.Quantity} \quad (4)$$

$$\text{Quantity} \sqsubseteq \exists \text{hasValue.xsd:double} \sqcap \exists \text{hasUnit.Unit} \quad (5)$$

We identified all units with classes from the Units Ontology (UO) [52]. Qualities are expressed using the Phenotype and Trait Ontology (PATO) [53] and linked to SIO as subclasses of the *Quality* class. To illustrate this pattern, we provide an examples represented in Turtle format. The example (Listing 1) demonstrates an environmental temperature measurement taken at a sampling site.

Listing 1 Turtle representation of an environmental temperature measurement.

```
rak:P000001 a rak:0000006 ; # Measurement Process
  sio:000291 rak:1000001 ; # hasTarget
  sio:000229 rak:4000001 . # hasOutput (Value)

rak:4000001 a rak:0000010 ; # Quantity
  sio:000221 uo:0000027 ; # hasUnit (degree Celsius)
  rak:2000003 "20.7"^^xsd:double ; # hasValue
  sio:000215 rak:5000001 . # refersTo (Quality)

rak:5000001 a pato:0000146 ; # Temperature Quality
  sio:000011 rak:1000001 . # isAttributeOf
```

In words, an instance **rak:P000001** of a measurement process has as its target the sampling site **rak:1000001** and produces as output the quantity **rak:4000001**. That quantity carries the literal value 20.7, is reported in degrees Celsius (**uo:0000027**), and refers to a temperature quality **rak:5000001** which is in turn an attribute of the same site. Reading the chain in either direction therefore connects the numeric reading to both the measured site and the abstract quality that it quantifies.

3.3.2 Sampling pattern

A sampling event is a process that derives a sample from a feature of interest (the site).

$$\text{SamplingProcess} \sqsubseteq \text{Process} \quad (6)$$

$$\text{Sample} \sqsubseteq \text{MaterialEntity} \quad (7)$$

$$\text{SamplingProcess} \sqsubseteq \exists \text{hasTarget.Site} \sqcap \exists \text{hasOutput.Sample} \quad (8)$$

$$\text{Sample} \sqsubseteq \exists \text{isDerivedFrom.Site} \quad (9)$$

Listing 2 illustrates the semantic representation of a surface soil sample collected during Trip 1.

Listing 2 Turtle representation of a sampling process.

```
rak:P100001 a rak:0000024 ; # Sampling Process
  sio:000291 rak:1000001 ; # hasTarget
  sio:000229 rak:7000001 . # hasOutput (Collection)

rak:6000001 a rak:0000021 ; # Deep Soil Sample
  sio:000252 rak:1000001 . # isDerivedFrom
```

Read in prose, an instance **rak:P100001** of a sampling process targets the site **rak:1000001** and yields a collection **rak:7000001** as its output; the deep soil sample **rak:6000001** is in turn asserted to be derived from the same site. The redundancy between **hasTarget** on the process and **isDerivedFrom** on the sample is intentional:

the former records the action, the latter records the resulting provenance edge directly on the material entity, so queries can traverse from sample to site without joining through the sampling event.

3.3.3 Experimental protocol pattern

We model experimental protocols as processes specified by a protocol entity. Each process (e.g., DNA extraction, library preparation) consumes inputs and produces outputs, explicitly linking agents and devices. The pattern instantiates SIO’s general *Process–hasInput/hasOutput–Object* idiom and adds a specification link to a reusable protocol description.

$$\text{ProtocolExecution} \sqsubseteq \text{Process} \quad (10)$$

$$\text{Protocol} \sqsubseteq \text{InformationContentEntity} \quad (11)$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{isSpecifiedBy}.\text{Protocol} \quad (12)$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{hasInput}.\text{MaterialEntity} \sqcap \exists \text{hasOutput}.\text{MaterialEntity} \quad (13)$$

Listing 3 demonstrates the formalization of a DNA extraction process.

Listing 3 Turtle representation of a DNA extraction protocol.

```
rak:P200001 a sio:000994 ; # Scientific Protocol Execution
  sio:000339 rak:DNA_ext_powersoil ; # isSpecifiedBy
  sio:000230 rak:6000001 ; # hasInput (Soil Sample)
  sio:000229 rak:6800001 . # hasOutput (DNA Extract)

rak:6800001 a rak:0000040 ; # DNA Extract
  sio:000008 rak:4900001 . # hasAttribute (Concentration Quality)
```

In words, the protocol-execution instance `rak:P200001` is specified by the named extraction protocol `rak:DNA_ext_powersoil`, takes the soil sample `rak:6000001` as input, and produces the DNA extract `rak:6800001` as output; the resulting extract carries a concentration quality (`rak:4900001`) which is itself quantified through a separate measurement process following the pattern of Equation 1.

3.3.4 Taxonomic abundance pattern

Taxonomic profiles represent collections of taxon observations containing associated relative and absolute abundances. Instead of modeling the feature table as a flat matrix, we formalize the entire table as a bioinformatic workflow that produces a dataset containing individual taxonomic observations. Here, we use the merged taxonomy (see Section 2) to assign specific NCBI or GTDB classes as the target of each observation. Listing 4 shows how we map an entire sample’s feature table row into the knowledge graph.

For instance, the sequence analysis for a specific sample yields an absolute sequence count associated with a taxonomic class from the NCBI Taxonomy. We construct an abundance quality that targets the explicit taxonomic identifier, binding the observational count back to the biological workflow. Formally, the pattern instantiates SIO’s *Collection–hasMember–Thing* and *Quantity–refersTo–Quality* idioms:

- BioinformaticWorkflow $\sqsubseteq$ Process (14)
- TaxonomicDataset $\sqsubseteq$ Collection (15)
- BioinformaticWorkflow $\sqsubseteq \exists \text{hasInput.SequenceDataset} \sqcap \exists \text{hasOutput.TaxonomicDataset}$ (16)
- TaxonomicDataset $\sqsubseteq \forall \text{hasMember.AbundanceObservation}$ (17)
- AbundanceObservation $\sqsubseteq \text{Quantity} \sqcap \exists \text{refersTo.AbundanceQuality}$ (18)
- AbundanceQuality $\sqsubseteq \exists \text{isAttributeOf.Taxon}$ (19)

Listing 4 Turtle representation of taxonomic abundance formalized from a feature table.

```

rak:P290001 a rak:0000071 ; # Bioinformatic Workflow
  sio:000230 rak:7900001 ; # hasInput (FASTQ Dataset)
  sio:000229 rak:7290001 . # hasOutput (Taxonomic Dataset)

rak:7290001 a rak:0000074 ; # Taxonomic Dataset
  sio:000059 rak:4400001 . # hasMember (Observation Value)

rak:4400001 a sio:000070 ; # Quantity (Absolute Count)
  sio:000215 rak:5400001 ; # refersTo
  rak:2000021 "15"^^xsd:integer . # absolute read count

rak:5400001 a rak:0000078 ; # Absolute Abundance Quality
  sio:000011 ncbitaxon:286 . # isAttributeOf (Pseudomonas)

```

Verbalized, the bioinformatic-workflow instance `rak:P290001` consumes a FASTQ dataset (`rak:7900001`) and produces the taxonomic dataset `rak:7290001`; that dataset has as one of its members the quantity `rak:4400001`, which records an absolute read count of 15 and refers to an absolute abundance quality that is in turn an attribute of *Pseudomonas* (`obo:NCBITaxon.286`). One row of a feature table is therefore expanded into a four-node sub-graph that preserves both the classifier output and its provenance.

3.3.5 XRF geochemistry pattern

We model XRF analysis as a process that targets a soil sample and produces multiple measurement values, each corresponding to the concentration of a specific chemical analyte. We map each analyte to the Chemical Entities of Biological Interest (ChEBI) ontology [37] as well as to the PubChem database [54]. Out of 93 monitored geochemical parameters, we mapped 80 analytes directly to corresponding ChEBI classes and aligned 92 parameters to PubChem Compound Identifiers. The XRF pattern is a specialization of the measurement pattern (Equation 1), with the target restricted to a soil sample and the produced quantity bound to an analyte-specific quality:

$$\text{XRFMeasurementProcess} \sqsubseteq \text{MeasurementProcess} \quad (20)$$

$$\text{XRFMeasurementProcess} \sqsubseteq \exists \text{hasTarget.SoilSample} \sqcap \exists \text{hasOutput.ConcentrationQuantity} \quad (21)$$

$$\text{ConcentrationQuantity} \sqsubseteq \text{Quantity} \sqcap \exists \text{refersTo}.\text{AnalyteQuality} \quad (22)$$

$$\text{AnalyteQuality} \sqsubseteq \exists \text{isAttributeOf}.\text{SoilSample} \quad (23)$$

Listing 5 illustrates an XRF measurement for a specific analyte.

Listing 5 Turtle representation of an XRF geochemistry measurement.

```
rak:P300001 a rak:0000080 ; # XRF Measuring Process
  sio:000291 rak:6000001 ; # hasTarget
  sio:000229 rak:4300001 . # hasOutput

rak:4300001 a sio:000070 ; # Quantity
  sio:000215 rak:5300001 ; # refersTo
  sio:000221 uo:0000187 ; # Percent unit
  rak:2000012 "58.2"^^xsd:double . # Concentration

rak:5300001 a rak:0000167 ; # Analyte Quality Class
  sio:000011 rak:6000001 . # isAttributeOf
```

In words, the XRF measurement process `rak:P300001` targets the soil sample `rak:6000001` and produces the quantity `rak:4300001`, which records a concentration of 58.2 percent (`uo:U0_0000187`) and refers to an analyte-specific quality `rak:5300001`. Because the analyte quality is typed with a dedicated subclass per element, the same pattern instance can be reused unchanged across all 91 geochemical analytes by varying only the quality class IRI.

3.3.6 Sampling site and biome assignment pattern

Each physical sampling location is represented as a persistent individual that is grounded in the Environment Ontology (ENVO) by linking to a biome and to one or more environmental features. We use both an existential and a universal restriction so that the asserted biome both holds and is the only admissible value, which preserves the closed-world intent of the field record.

$$\text{SamplingSite} \sqsubseteq \text{Site} \quad (24)$$

$$\text{SamplingSite} \sqsubseteq \exists \text{hasBiome}.\text{ENVO:Biome} \sqcap \forall \text{hasBiome}.\text{ENVO:Biome} \quad (25)$$

$$\text{SamplingSite} \sqsubseteq \exists \text{hasEnvironmentalFeature}.\text{ENVO:EnvironmentalFeature} \quad (26)$$

Listing 6 Turtle representation of a sampling site grounded in ENVO.

```
rak:1000001 a rak:0000002 ; # Sampling Site
  rdfs:label "Site 11" ;
  geo:asWKT "POINT(50.123 22.456)"^^geo:wktLiteral ;
  rak:2000050 envo:01000219 ; # hasBiome (desert biome)
  rak:2000051 envo:00000086 . # hasEnvironmentalFeature (sand dune)
```

In words, the sampling site `rak:1000001` carries a label, a WKT geometry, and is asserted both to belong to the ENVO desert biome and to exhibit a sand-dune environmental feature. The combined existential and universal restriction on `hasBiome` (Equation 24) ensures that any reasoner treats the asserted biome as the only admissible value for that site.

3.3.7 Sequencing workflow pattern

Sequencing data are produced by a chain of laboratory and computational processes that the knowledge graph represents as a sequence of *Process–hasInput/hasOutput–Object* steps: a DNA extraction process consumes a soil sample and produces a DNA extract; a library preparation process consumes the DNA extract and produces an amplicon library; a sequencing process consumes the amplicon library and produces a FASTQ dataset whose members are sequence reads. Each process is specified by a protocol and may carry agents and device participants, reusing the experimental protocol pattern of Equation 10.

$$\text{DNAExtractionProcess} \sqsubseteq \text{ProtocolExecution} \quad (27)$$

$$\text{DNAExtractionProcess} \sqsubseteq \exists \text{hasInput}.\text{SoilSample} \sqcap \exists \text{hasOutput}.\text{DNAExtract} \quad (28)$$

$$\text{LibraryPreparationProcess} \sqsubseteq \text{ProtocolExecution} \quad (29)$$

$$\text{LibraryPreparationProcess} \sqsubseteq \exists \text{hasInput}.\text{DNAExtract} \sqcap \exists \text{hasOutput}.\text{AmpliconLibrary} \quad (30)$$

$$\text{SequencingProcess} \sqsubseteq \text{ProtocolExecution} \quad (31)$$

$$\text{SequencingProcess} \sqsubseteq \exists \text{hasInput}.\text{AmpliconLibrary} \sqcap \exists \text{hasOutput}.\text{FASTQDataset} \quad (32)$$

$$\text{FASTQDataset} \sqsubseteq \text{Dataset} \sqcap \forall \text{hasMember}.\text{SequenceRead} \quad (33)$$

Listing 7 Turtle representation of the sequencing workflow from soil sample to FASTQ dataset.

```
rak:P200001 a rak:0000044 ; # DNA Extraction Process
  sio:000339 rak:L000001 ; # isSpecifiedBy (PowerSoil protocol)
  sio:000230 rak:6000001 ; # hasInput (Soil Sample)
  sio:000229 rak:6800001 . # hasOutput (DNA Extract)

rak:P250001 a rak:0000045 ; # Library Preparation Process
  sio:000339 rak:L000010 ; # Illumina 16S Prep Guide
  sio:000230 rak:6800001 ; # hasInput (DNA Extract)
  sio:000229 rak:6900001 . # hasOutput (Amplicon Library)

rak:P280001 a rak:0000046 ; # Sequencing Process
  sio:000230 rak:6900001 ; # hasInput (Amplicon Library)
  sio:000229 rak:7900001 . # hasOutput (FASTQ Dataset)

rak:7900001 a rak:0000064 ; # FASTQ Dataset
  rdfs:seeAlso <https://www.ebi.ac.uk/ena/browser/view/ERR1234567> .
```

Read as a chain, the DNA extraction process `rak:P200001` consumes the soil sample `rak:6000001` and produces the DNA extract `rak:6800001`; that extract is the input of the library preparation `rak:P250001`, which yields the amplicon library `rak:6900001`; finally the sequencing process `rak:P280001` consumes the library and produces the FASTQ dataset `rak:7900001`, which is cross-referenced to its ENA accession via `rdfs:seeAlso`. Each link in the chain is the same `hasInput/hasOutput` pattern, so a single SPARQL traversal can recover the full provenance from any node.

3.3.8 Sequencing quality control pattern

Sequencing quality control is represented as an additional computational process that takes a FASTQ dataset as input and produces multiple measurement values, each instantiating the generic measurement pattern of Equation 1. Metrics (sequence count, GC content, duplicate rate, read length) are split into forward (R1) and reverse (R2) subclasses so that the provenance of paired-end runs is preserved.

$$\text{SequencingQCProcess} \sqsubseteq \text{Process} \quad (34)$$

$$\text{SequencingQCProcess} \sqsubseteq \exists \text{hasInput.FASTQDataset} \sqcap \exists \text{hasOutput.QCMeasurementValue} \quad (35)$$

$$\text{QCMeasurementValue} \sqsubseteq \text{Quantity} \sqcap \exists \text{refersTo.QCQuality} \quad (36)$$

$$\text{QCQuality} \sqsubseteq \exists \text{isAttributeOf.FASTQDataset} \quad (37)$$

Listing 8 Turtle representation of a sequencing QC measurement (forward-read GC content).

```

rak:P350001 a rak:0000047 ; # Sequencing QC Process
  sio:000230 rak:7900001 ; # hasInput (FASTQ Dataset)
  sio:000229 rak:4500001 . # hasOutput (QC Value)

rak:4500001 a rak:0000086 ; # GC Content Value
  sio:000215 rak:5500001 ; # refersTo
  sio:000221 uo:0000187 ; # Percent
  rak:2000073 "54.2"^^xsd:double . # has GC content

rak:5500001 a rak:0000074 ; # GC Content Quality (Forward)
  sio:000011 rak:7900001 . # isAttributeOf (FASTQ Dataset)

```

In words, the QC process `rak:P350001` takes the FASTQ dataset `rak:7900001` as input and produces the measurement value `rak:4500001`, which records a GC content of 54.2 percent and refers to a forward-read GC-content quality `rak:5500001` that is in turn an attribute of the same FASTQ dataset. Splitting the quality class into forward and reverse subclasses preserves the read direction, so that paired-end QC metrics can be queried independently per orientation.

3.3.9 Laboratory-control pattern

A laboratory control is an occurrence-specific material entity, not a property of an expedition and not the string used to label a library. Each physical control carries its own instance of a positive, negative, extraction-blank or PCR-blank role, and a laboratory process realizes that role while consuming the material as input and target and producing a derived material or dataset. Processes are grouped into batches, and, because an extraction day can contain samples from several expeditions, neither the control nor its batch is linked directly to a trip.

$$\text{ControlMaterial} \sqsubseteq \text{MaterialEntity} \quad (38)$$

$$\text{ControlRole} \sqsubseteq \text{Role} \quad (39)$$

$$\text{ControlMaterial} \sqsubseteq \exists \text{hasRole}.\text{ControlRole} \quad (40)$$

$$\begin{aligned} \text{LaboratoryProcess} \sqsubseteq & \exists \text{hasInput}.\text{ControlMaterial} \sqcap \exists \text{hasTarget}.\text{ControlMaterial} \\ & \sqcap \exists \text{realizes}.\text{ControlRole} \end{aligned} \quad (41)$$

$$\text{LaboratoryProcess} \sqsubseteq \exists \text{isPartOf}.\text{ProcessingBatch} \quad (42)$$

Listing 9 instantiates the pattern for an extraction blank. The blank material **rak:C000001** bears an extraction-blank role **rak:R000001** through **sio:000228** (*has role*); the extraction process **rak:P400001** takes the blank as input and target, realizes the role through **sio:000355**, and is part of the extraction batch **rak:B000001**. Asserting the role as an individual rather than as a class membership keeps each physical control distinct even when labels are reused across expeditions.

Listing 9 Turtle representation of an extraction-blank control and its role realization.

```
rak:C000001 a rak:0000300 ; # Extraction Blank Material
sio:000228 rak:R000001 . # hasRole

rak:R000001 a rak:0000301 ; # Extraction Blank Role
sio:000227 rak:C000001 . # isRoleOf

rak:P400001 a rak:0000302 ; # Extraction Process
sio:000230 rak:C000001 ; # hasInput
sio:000291 rak:C000001 ; # hasTarget
sio:000355 rak:R000001 ; # realizes
sio:000068 rak:B000001 . # isPartOf (Batch)

rak:B000001 a rak:0000310 . # Extraction Batch
```

Uncertain identity, batch membership and assay applicability are represented as evidence-bearing statements with an explicit confirmed, provisional, unresolved or inapplicable status rather than as ground truth, so a reused or ambiguous label never merges two physical controls. Whole-cell D6300 and purified-DNA D6322 controls enter the workflow at different processes, which lets the graph distinguish extraction and lysis evaluation from PCR, library and sequencing evaluation. The control module contains 489 entities and 3,800 triples, including 51 material-specific role individuals and 47 sequence occurrences.

3.3.10 Archived-soil pH pattern

Archived-soil pH instantiates the generic measurement pattern of Equation 1 but binds it to an archived specimen rather than to a field site. The measuring process consumes and targets an existing specimen, outputs a single pH value, is specified by the CaCl_2 protocol, and is part of a dated measurement session. The value is the measurement value of an acidity quality borne by the same specimen.

$$\begin{aligned} \text{pHMeasurementProcess} \sqsubseteq & \text{MeasurementProcess} \sqcap \exists \text{hasInput}.\text{Specimen} \\ & \sqcap \exists \text{hasTarget}.\text{Specimen} \sqcap \exists \text{hasOutput}.\text{pHValue} \end{aligned} \quad (43)$$

$$\text{pHValue} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf}.\text{AcidityQuality} \quad (44)$$

$$\text{AcidityQuality} \sqsubseteq \text{Quality} \sqcap \exists \text{isAttributeOf.Specimen} \quad (45)$$

$$\text{pHMeasurementProcess} \sqsubseteq \exists \text{isPartOf.MeasurementSession} \quad (46)$$

Listing 10 shows a single admitted pH observation. The specimen `rak:6000001` bears the acidity quality `rak:5600001`; the value `rak:4600001` carries the literal 7.8, has unit `UO:0000196`, and points back to the quality with `sio:000215` and to its producing process with `sio:000232`. The process is specified by the `CaCl2` protocol and is part of the session `rak:B100001`. Rows that fail source, date, specimen or quality-control admission remain in the audit table and are not asserted as measurements (Section 2.7).

Listing 10 Turtle representation of an archived-soil pH measurement.

```
rak:P500001 a sio:001054 ; # Measuring Process
  sio:000230 rak:6000001 ; # hasInput (Specimen)
  sio:000291 rak:6000001 ; # hasTarget
  sio:000229 rak:4600001 ; # hasOutput (pH Value)
  sio:000339 rak:L000020 ; # isSpecifiedBy (CaCl2 Protocol)
  sio:000068 rak:B100001 . # isPartOf (Session)

rak:4600001 a sio:001089 ; # pH Value
  sio:000300 "7.8"^^xsd:double ; # hasValue
  sio:000221 uo:0000196 ; # hasUnit (pH)
  sio:000215 rak:5600001 ; # isMeasurementValueOf
  sio:000232 rak:P500001 . # isOutputOf

rak:5600001 a pato:0001842 ; # Acidity Quality
  sio:000011 rak:6000001 ; # isAttributeOf (Specimen)
  sio:000216 rak:4600001 . # hasMeasurementValue
```

3.4 OWL and RDF implementation

We generated the OWL/RDF implementation using custom Groovy scripts utilizing the OWL API 5 [55]. The knowledge base employs a modular design, separating the base ontology, sites, samples, measurements, XRF data, DNA extraction records, sequencing metadata, quality control metrics, laboratory controls, archived-soil pH, and taxonomic abundances into individual ABox modules [56].

We use OpenLink Virtuoso [57] for RDF storage and query answering. In a development benchmark, a project-specific forward-rule program applied selected RDFS/OWL rules for subclass, subproperty, domain, range, inverse, symmetric, equivalence and transitive-property assertions. The program used three scheduled passes, including only a one-hop transitive pass. It did not demonstrate convergence to a fixed point and did not implement arbitrary OWL property chains. We therefore describe this output as a *selected rule expansion*, not as complete OWL-Horst [58] closure, and we report exact module-level entity counts (Section 4) rather than historical asserted or inferred whole-graph triple counts.

3.5 Schema definition

We used Shape Expressions (ShEx) [48] to define structural schemas for the core entity types. While OWL provides open-world reasoning and inference capabilities, it

cannot readily enforce data completeness constraints. We developed ShEx schemas to explicitly validate the graph’s structure, ensuring data quality and compliance before loading it into the database.

Our ShEx implementation defines shapes for Sampling Sites, Soil Samples, Environmental Measurements, DNA Extracts, XRF processes, Sequencing Datasets, laboratory controls, and archived-soil pH observations. These shapes guarantee that essential attributes are invariably present. For instance, Listing 11 demonstrates how the schema enforces that every Sampling Site is explicitly typed and labeled, while capturing geographic coordinates and an optional textual description.

Listing 11 ShEx shape for Sampling Site validation.

```
<#SamplingSiteShape> {
  rdf:type [rak:0000002] ; # Must be of type Sampling Site
  rdf:type . * ; # May have other inferred types
  rdfs:label . ; # Must have a label
  dcterms:description xsd:string ? ; # Optional description
  geo:asWKT . ; # Mandatory geographic coordinates
}
```

In words, the shape `<#SamplingSiteShape>` requires that every conforming node carry an explicit type assertion to `rak:0000002` (Sampling Site) and an `rdfs:label`; it permits any number of additional inferred types and optionally allows a free-text description. Coordinates are mandatory in the shape, and all 70 sampling-site individuals in the released module satisfy it. Records that omit the mandatory type, label or geometry are rejected before they reach the triple store, and a deliberately coordinate-free fixture in the negative-fixture suite (Section 5) proves that the rejection fires.

The remaining shapes reuse the same closed-world idiom but encode the domain-specific provenance and measurement contracts that a type assertion alone cannot express. Listing 12 shows the soil-sample shape: a conforming sample must be explicitly typed as a replicate, carry a label and a stable source identifier, and be derived from a sampling site. The derivation edge is what lets a query traverse from a specimen to its site without joining through the sampling event.

Listing 12 ShEx shape for a soil sample.

```
<#SampleShape> {
  rdf:type [sio:001418] ; # Replicate
  rdf:type IRI * ; # Plus a specific material type
  rdfs:label xsd:string ;
  dc:identifier xsd:string ; # Stable source identifier
  sio:000244 IRI ; # isDerivedFrom -> Site
}
```

The DNA-extract and XRF shapes enforce the canonical measurement reification (split quality and value) of Equation 1. Listing 13 requires the concentration value to carry a numeric concentration, the nanograms-per-microlitre unit, and links back to both its quality and its producing measurement process, so a value that is detached from its process or that uses the wrong unit is rejected.

Listing 13 ShEx shape for a DNA-concentration value.

```

<#DNAConcentrationValueShape> {
  rdf:type [rak:0000044] ;      # Concentration value
  sio:000215 @<#DNAConcentrationQualityShape> ; # -> Quality
  sio:000232 @<#MeasurementProcessShape> ;      # -> Process
  sio:000221 [uo:0000275] ;      # hasUnit -> ng/uL
  rak:2000015 xsd:double ;      # hasDNAConcentration
}

```

The XRF shape encodes the two acquisition workflows directly: a conforming XRF process must either take a collected specimen as input (the laboratory workflow) or target a recorded sampling site (the field workflow), and it must produce at least one value. This disjunction is what allows one shape to cover both workflows while still rejecting a process that has neither an input nor a target, the defect exercised by one of the negative fixtures.

Listing 14 ShEx shape for an XRF measurement process.

```

<#XRFProcessShape> {
  rdf:type [rak:0000025] ;
  (
    sio:000230 IRI |      # lab: hasInput -> specimen
    sio:000291 IRI      # field: hasTarget -> site
  ) ;
  sio:000339 IRI ;      # isSpecifiedBy -> Protocol
  sio:000229 @<#XRFValueShape> + ; # hasOutput -> 1+ values
}

```

4 Data records

The resource is distributed as a single directory tree in which every payload file is declared in a machine-readable manifest (`PRE_RELEASE_MANIFEST.tsv`) with its byte size, SHA-256 checksum, category, release status and licence gate. The raw sequencing data are deposited in the European Nucleotide Archive (ENA) under the umbrella project accession PRJEB104209, with amplicon sequencing under project PRJEB106069. The knowledge graph is accessible via a public SPARQL endpoint at <https://rubalkhali.science/sparql> and through a web interface at <https://rubalkhali.science/>, and all source code for data processing, ontology generation, and validation is maintained at <https://github.com/bio-ontology-research-group/empty-quarter>.

4.1 Knowledge-graph modules

The primary artifacts are the knowledge-graph modules, distributed as OWL/XML and Turtle files and structured by data type (Table 3). The base ontology (`rubalkhali.owl`) holds the terminology; each remaining module holds the instance data for one domain and shares the same site, specimen and process identifiers, so the modules can be loaded together into any RDF store or used individually.

4.2 Tabular data products

Alongside the graph, the deposit distributes the curated tabular data products from which the modules are generated: the versioned source ledger linking site and trip to

Table 3 Knowledge-graph modules in the deposit.

Module	Content
<code>rubalkhali.owl</code>	Base ontology (TBox)
<code>rubalkhali_sites.owl</code>	Sampling sites with coordinates and biome annotations
<code>rubalkhali_samples.owl</code>	Soil and plant specimen individuals with compartment types
<code>rubalkhali_measurements.owl</code>	Field environmental measurements and climate records
<code>rubalkhali_xrf.owl</code>	XRF measurement values from the field and laboratory workflows
<code>rubalkhali_dna.owl</code>	DNA extracts with concentration data
<code>rubalkhali_sra.owl</code>	Sequencing run records linked to ENA accessions
<code>rubalkhali_qc.owl</code>	Sequencing quality metrics (GC content, duplicate rate, sequence counts)
<code>rubalkhali_controls.ttl</code>	Laboratory-control materials, roles, batches and sequence occurrences
<code>rubalkhali_ph_eq_ph_shared.v1.0.0.ttl</code>	Archived-soil pH measurements and sessions
<code>rubalkhali_taxonomy_abox.ttl</code>	Taxon abundance observations over the feature profiles

specimen, extraction, library, sequence run, quality-control decision and derived community; the canonical Trips 1–5 amplicon feature table and its taxonomy mapping; the correction and alias ledgers that record every deterministic metadata repair without overwriting source values; the curated field environmental table; the field and laboratory XRF tables together with the audited chemical-mapping ledger; and the versioned archived-soil pH dataset EQ-PH-SHARED-v1.0.0. The four derived companion-analysis inputs (the paired PMA table, the CoverM genome-abundance profiles, the eggNOG annotations and the measured-function input archive) are distributed as records for downstream statistical re-execution rather than as graph modules.

4.3 Validation schemas, queries and evidence

The deposit includes the artifacts needed to re-validate the resource: the ShEx shapes that define the structural contract for each entity family, the positive and negative structural fixtures, the archived competency SPARQL queries, and the machine-readable evidence bundles (reconciliation summaries, audit tables, validation records and checksum manifests) produced by the release workflow. A companion data dictionary (`metadata/DATA_DICTIONARY.tsv`) records the field names, data types, units, missing-value conventions and definitions for the curated canonical tabular records. Quantities carry units only where a unit is documented at the source: environmental source tables retain degrees Celsius, hectopascals and percent relative humidity; in the graph, pressure magnitudes are asserted in pascals, climate records use degrees Celsius and millimetres, and DNA concentrations are in nanograms per microlitre.

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XRF concentrations carry no documented unit assertion and must not be treated as percentages.

The data and source code to generate the Rub' al-Khali knowledge graph is available at .

5 Technical validation

We validated the knowledge graph using four complementary approaches, each targeting a distinct aspect of data quality: competency question evaluation via SPARQL queries, structural validation using Shape Expressions (ShEx), logical consistency checking with the ELK reasoner, and usability assessment through a web portal.

5.1 Competency questions and SPARQL queries

To validate whether the full knowledge graph can be used to answer queries about the Rub' al-Khali, we defined competency questions expressed as SPARQL queries [59] with known expected answers. We automated this validation process to enable regression testing during ontology updates. The validated competency questions include retrieving XRF measurements, listing sampling sites with coordinates, extracting DNA extraction data, tracing provenance from soil samples to sequencing runs, and querying taxonomic composition. The competency questions are designed to test questions that are needed for operating the website, and to support different types of queries within our dataset.

Table 4 lists the competency questions we predefined for validation. Each is expressed as a SPARQL query with a known expected answer and is run as a regression test whenever the knowledge graph is regenerated. Together they cover every core module: geochemistry (CQ1, CQ4), the site catalogue and its biome grounding (CQ2, CQ3), the molecular provenance chain (CQ5, CQ6), and taxonomic composition (CQ7).

Table 4 Predefined competency questions used for validation.

ID	Competency question
CQ1	What are the XRF analyte concentrations measured at a given sampling site?
CQ2	Which sampling sites are recorded, and what are their GPS coordinates?
CQ3	Which biome and environmental features are assigned to each sampling site?
CQ4	What are the Light Elements percentages across the XRF measurements?
CQ5	Which DNA extraction records (extract, concentration, protocol) exist for a given sample?
CQ6	What is the full provenance chain from a soil sample to its sequencing run?
CQ7	What is the taxonomic composition and abundance of a given sample or run?

We show several example SPARQL queries. Listing 15 retrieves XRF measurements for a specific site, requiring traversal of the measurement process, quality, and value entities.

Listing 15 SPARQL query to retrieve field XRF measurements for Site 10. The query traverses from sample to measurement process to individual analyte concentrations.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?processLabel ?analyte ?concentration ?unitLabel
WHERE {
  ?sample a rak:0000021 ;
          rdfs:label ?sampleLabel .
  FILTER(REGEX(?sampleLabel, "Site 10"))
  ?value sio:000232 ?process .
  ?process rdfs:label ?processLabel .
  FILTER(REGEX(?processLabel, "Field", "i"))
  ?sample sio:000008 ?quality .
  ?quality sio:000215 ?value .
  ?value rak:2000012 ?concentration .
  ?value sio:000221 ?unit .
  ?unit rdfs:label ?unitLabel .
  ?quality a ?qualityClass .
  ?qualityClass rdfs:label ?analyte .
  FILTER(?qualityClass != rak:0000029)
}
ORDER BY ?processLabel ?analyte LIMIT 100
```

Read in prose, the query first selects every soil sample whose label matches Site 10, then walks back from each measurement value to its producing process and forwards from the sample through its quality and value to the recorded concentration and unit. The final filter excludes the Light Elements pseudo-analyte (rak:0000029) so that only individually quantified elements appear in the result, and the query returns one row per process and analyte combination.

Listing 16 retrieves taxonomic abundance profiles, joining across bioinformatic workflows, FASTQ datasets, and taxon observations.

Listing 16 SPARQL query to retrieve taxonomic composition and abundance for all samples.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT DISTINCT ?run ?taxonLabel ?count ?relativeAbundance
WHERE {
  ?proc a rak:0000071 ;
        sio:000230 ?fastq ;
        sio:000229 ?dsAbs , ?dsRel .
  ?dsAbs a rak:0000074 .
  ?dsRel a rak:0000075 .
  ?fastq rdfs:label ?runLabel .
  BIND(REPLACE(?runLabel, "FASTQ dataset for ", "") AS ?run)
  ?dsAbs sio:000059 ?valAbs .
  ?valAbs rak:2000021 ?count .
  ?valAbs sio:000215 ?qualAbs .
  ?qualAbs sio:000011 ?taxon .
  ?taxon rdfs:label ?taxonLabel .
  ?qualRel sio:000011 ?taxon ;
           a rak:0000072 ;
           sio:000214 ?valRel .
  ?valRel rak:2000020 ?relativeAbundance .
  FILTER(?taxon != ?fastq)
}
ORDER BY ?run DESC(?count) LIMIT 100
```

In words, the query identifies every bioinformatic workflow that produces both an absolute and a relative taxonomic dataset for the same FASTQ input, then for each absolute observation it retrieves the read count, the associated taxon label, and the matching relative abundance from the parallel relative dataset. Results are grouped by sequencing run and ordered by descending count, so the most abundant taxa per run appear first.

We added an additional query (Listing 17) to retrieve specific taxa abundances alongside environmental constraints, demonstrating the ability to cross-reference biological and environmental properties.

Listing 17 SPARQL query to retrieve *Pseudomonas* relative abundance at sites with high temperature.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX pato: <http://purl.obolibrary.org/obo/PATO_>
PREFIX ncbitaxon: <http://purl.obolibrary.org/obo/NCBITaxon_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?siteLabel ?temp ?relAbundance
WHERE {
  ?site a rak:0000002 ;
        rdfs:label ?siteLabel .

  # Environmental Temperature
  ?tempProcess sio:000291 ?site ;
               sio:000229 ?tempVal .
  ?tempVal rak:2000012 ?temp ;
           sio:000215 ?tempQual .
  ?tempQual a pato:0000146 .
  FILTER(?temp > 35.0)

  # Derived Sample and Taxon Abundance
  ?sample sio:000252 ?site .
  ?ext sio:000230 ?sample ; a rak:0000041 ; sio:000229 ?dna .
  ?libPrep sio:000230 ?dna ; a rak:0000062 ; sio:000229 ?lib .
  ?seq sio:000230 ?lib ; a rak:0000064 ; sio:000229 ?fastq .

  ?bioinf a rak:0000071 ;
          sio:000230 ?fastq ;
          sio:000229 ?dsRel .
  ?dsRel a rak:0000075 ;
         sio:000059 ?valRel .
  ?valRel rak:2000020 ?relAbundance ;
         sio:000215 ?qualRel .
  ?qualRel sio:000011 ncbitaxon:286 . # Pseudomonas
}
```

In words, the query first selects every sampling site at which the recorded environmental temperature exceeds 35 °C, then for each such site it walks the full provenance chain — soil sample, DNA extract, library preparation, sequencing run, FASTQ dataset, bioinformatic workflow, relative abundance dataset — down to the relative abundance value of *Pseudomonas* (ncbitaxon:286). Each row of the result therefore reports a single hot site together with its corresponding *Pseudomonas* relative abundance, demonstrating a join that crosses the environmental, geochemical, and biological domains in a single SPARQL query.

5.2 Source reconciliation and measurement validation

The executed evidence workflow produced a row-level sample ledger and reconciliation summaries for specimens, extracts, libraries, runs and community profiles. The 2,516 eligible source rows reconcile exactly to 2,516 generated sample individuals; the 1,647 extraction-eligible rows reconcile to 1,647 DNA extracts; and 1,242 resolved sequence rows reconcile to 1,242 amplicon libraries and 1,242 FASTQ datasets. A site-alias gate independently matches each Trip 1 numeric identifier 61–64 to a named catalogue individual by exact coordinates and rejects coordinate or IRI disagreement, accounting for all 36 affected source rows.

The environmental-metadata gate reproduced 274 nonblank source records and verified all 24 exact-cell correction-ledger dispositions against the corresponding raw cells. All retained field temperatures, pressures and relative humidities fell within their declared physical ranges; the unresolved Trip 5 site 40 value of 194% was preserved in the ledger but is absent from both the curated table and the environmental RDF module. The XRF provenance audit reproduced all 13,390 reported Trips 1–4 cells under the maximum-positive rule and all 4,293 reported Trip 5 cells under the last-reported rule, with no mismatched reported cells, and verified that the canonical laboratory input comprises all 725 processed records. An independent chemical-identifier gate checked all 93 instrument channels against the pinned ChEBI v247 release and a checksummed PubChem snapshot; all rows passed, and audited nulls are retained for the unmapped LE pseudo-analyte and unsupported oxides rather than being replaced by lexical matches. The pH reconciliation independently accounts for all 1,168 workbook rows, matches every admitted row to exactly one existing specimen, and reproduces the five dispositions reported in Section 4.

5.3 ShEx schema validation

We validated the structural integrity of the knowledge graph using Shape Expressions (ShEx), as described in Section 3.5. We defined ShEx shapes for all core entity types: sampling sites, soil samples, environmental measurements, DNA extracts, sequencing datasets, quality control records, and taxonomy observations. We bound each shape to its corresponding class instances via a Shape Map.

We implemented the validation using the Apache Jena ShEx library (version 4.10.0) invoked from Groovy scripts. We integrated ShEx validation into the automated deployment workflow: each time the workflow regenerates an ABox module, the validation step tests instances of the relevant class against their ShEx shape before loading the data into Virtuoso. The workflow terminates with an error if any instance violates the shape constraints (e.g., a sampling site lacking GPS coordinates or a DNA extract missing a concentration value). This mechanism functions as a regression test, ensuring that modifications to the ETL scripts do not introduce structural inconsistencies. The pH module was validated separately over all 712 measurement processes, values and acidity qualities and all 29 sessions, and the laboratory-control module passed its shape and scientific-invariant checks.

The tractable modules passed their assigned ShEx shapes. The ShEx suite also tests rejection, not only conformance: negative fixtures that differ from a valid instance

in exactly one respect (a sampling site without coordinates, a value with an invalid datatype, a reversed value-to-quality direction, and an XRF process with neither an input nor a target) are each rejected by the released shapes, so a passing negative test identifies the constraint that caught it. The multi-gigabyte taxonomy ABox remains outside this memory-bounded run; it is instead parsed in full by an independent Turtle parser and scanned by a streaming structural validator, which accounted for 44,528,482 triples in 5,953,690 subject blocks with zero structural violations. This establishes full-file syntax and generator-specific structural validation, not a claim that the taxonomy ABox passed the unexecuted ShEx shapes.

5.4 Logical consistency

We verified the logical consistency of the knowledge graph using the ELK reasoner, as described in Section 2. We loaded the tractable set of OWL modules into a single OWL ontology and ran ELK to check for unsatisfiable classes and contradictory axioms. The reasoner reported the merged included modules as consistent.

We performed this consistency check as part of continuous integration during ontology development. Whenever we modified the TBox or introduced new ABox modules, we re-ran the ELK reasoner to detect regressions. We chose ELK over more expressive reasoners such as Hermit [50] because the size of the knowledge graph made reasoning in the full OWL 2 DL profile computationally infeasible.

We implemented the ELK consistency check via the OWL API (version 4.5.26) with the ELK OWL API module (version 0.4.3) in a Groovy script. A clean rerun on the current tractable modules loaded 781,293 axioms from 10 distinct ontology IRIs and reported the resulting merged ontology as consistent; the generated ecosystem and laboratory-control modules were included. The run excluded the multi-gigabyte taxonomy ABox and large reference modules and did not precompute the class hierarchy, so it establishes bounded consistency of the included modules rather than whole-graph OWL 2 DL consistency. ELK is incomplete for universal restrictions, inverse properties and other constructs used by imported modules, and the excluded ABoxes were not covered; likewise, the selected forward-rule program of Section 3.4 has not been demonstrated to reach a fixed point and is not complete OWL-Horst closure [58].

5.5 Compute-time metrics

Table 5 summarizes the computational cost of the key deployment steps of the reference Virtuoso deployment. The selected forward-rule expansion (three passes; see Section 3.4) runs in approximately 114 minutes. ELK consistency checking over the asserted modules takes under 2 seconds. Competency question latency on the Virtuoso endpoint ranges from 5 ms for simple site look-ups (CQ2) to over 1 second for queries that traverse inverse properties across the entire DNA extraction workflow (CQ5).

The taxonomy composition query (CQ7) exceeds a 5-minute timeout because it joins absolute and relative abundance datasets across 1.4 million observations. This query is not used by the web portal; the portal retrieves taxonomic data per sample rather than across all runs, which completes in sub-second time.

Table 5 Compute-time metrics for key deployment and validation steps.

Step	Time
RDF module loading (Virtuoso, 9 ABox files)	140 s
Selected rule expansion (3 passes)	114 min
ELK classification (asserted only)	1.6 s
CQ1 – XRF measurements, Site 10	23 ms
CQ2 – Sampling sites + GPS	5 ms
CQ3 – Biomes per site	5 ms
CQ4 – Light Elements percentages	8 ms
CQ5 – DNA extraction records	1,342 ms
CQ6 – Provenance chain	255 ms
CQ7 – Taxonomic composition	>300 s

5.6 Usability validation (web portal)

We developed a web portal (<https://rubalkhali.science/>) to provide user-friendly access to the data. The application architecture provides an end-to-end validation of the knowledge graph: the backend dynamically generates SPARQL queries to fetch content directly from the knowledge graph in real time without maintaining a separate relational database. Every page rendered constitutes a functional test of the underlying data. This architecture ensures that the website always reflects the current state of the knowledge graph, and any data integrity problem manifests as a visible error on the portal.

5.7 FAIR compliance

The design decisions described in the preceding sections collectively implement the FAIR Data Principles [27]. We describe how the resource addresses each principle in turn.

To make the data *findable*, we assigned a globally unique and persistent IRI under the `https://rubalkhali.science/kb/` namespace to every named individual, so that each site, specimen, process, value and dataset is directly and stably addressable rather than identified only by a row in a table. The sequencing data are registered with the European Nucleotide Archive under the umbrella accession PRJEB104209, and the resource is described with rich metadata: the machine-readable manifest indexes every payload file, and the data dictionary defines every curated field. Because the knowledge graph encodes its own terminology, the metadata and the data share one namespace and can be retrieved together.

To make the data *accessible*, we expose the knowledge graph through a standard SPARQL 1.1 endpoint and a web interface, both of which use open, universally implementable protocols and require no authentication for read access. The raw sequences are retrievable from ENA through their accessions, and the deposited modules can be downloaded and loaded into any RDF store, so access does not depend on the continued operation of our endpoint.

To make the data *interoperable*, we represent all content with established shared vocabularies: SIO as the upper-level framework, ENVO for environmental context, ChEBI for chemical entities, PATO for qualities, UO for units, and NCBI Taxonomy for taxa. The graph uses the standard formal languages OWL 2 DL, RDF/Turtle and SPARQL, and because domain entities are grounded in these widely used ontologies, the resource can be joined with other datasets that use the same vocabularies. The qualified references between modules (for example from a specimen to its site or from a value to its producing process) are themselves asserted as RDF, so the cross-references that give the data their meaning travel with the data.

To make the data *reusable*, we describe the resource with detailed and accurate provenance that records the complete chain from physical sample collection through DNA extraction, library preparation, sequencing and quality control to the derived community profiles, together with the correction and alias ledgers that document every deterministic repair to the source records. The data are released under CC-BY 4.0, the generation and validation code is open-sourced, and the release is versioned and archived so that a cited version remains stable. These attributes let a reuser understand the origin and limitations of each observation and integrate the resource into new work with confidence.

6 Usage notes

The knowledge graph is publicly accessible via the SPARQL endpoint at <https://rubalkhali.science/sparql>. Users can submit SPARQL 1.1 queries using standard clients (e.g., `curl`, Python’s `SPARQLWrapper`, R’s `SPARQL` package). The web portal at <https://rubalkhali.science/> provides a graphical interface for browsing sites, samples, and measurements without requiring SPARQL expertise.

The semantic structure of the knowledge graph enables cross-domain queries. For example, researchers investigating the relationship between soil geochemistry and microbial diversity can construct a single query that retrieves calcium concentrations alongside Shannon diversity indices computed from taxon abundances. Executing this query in a relational database would necessitate joining at least four separate tables.

The modular OWL architecture permits extending the knowledge graph with new data types, such as metabolomics or remote sensing data. Users can introduce new classes as subclasses of existing SIO entities and generate corresponding ABox modules. We automate the entire lifecycle—from data generation through validation to database loading—using the provided deployment workflow (`deploy_onto.sh`).

6.1 Design rationale

We adopted the Semanticscience Integrated Ontology (SIO) as the upper-level framework because it offers a small set of relations (`hasInput`/`hasOutput`, `hasTarget`, `hasMember`, `refersTo`, `hasValue`) that generalize across sampling, measurement, and provenance patterns. Sample collection, XRF analysis, DNA extraction, sequencing, and taxonomic classification are all instantiations of the same `Process--hasInput/hasOutput--Object` idiom; quantitative results are uniformly

represented as `Quantity refersTo Quality` pairs with explicit units. This uniformity enables cross-domain SPARQL queries that traverse from a climate observation through a sampling site to a taxon abundance without per-domain join logic. Recent work on simplified upper-level ontologies, such as SULO [60], takes this argument further by strengthening fundamental relations and reducing reliance on specialized predicates; our patterns are compatible with this direction.

We chose ShEx for ingest validation because the ETL is batch-oriented and the validator runs in the same Groovy/Jena toolchain that produces the RDF. SHACL is an obvious alternative supported in-database by several triple stores; a future port of the shapes from ShEx to SHACL would enable in-database validation at load time. We deployed Virtuoso as the triple store because it is available as free software under the GNU General Public License, supports OWL-Horst forward-chaining materialization, and has demonstrated operational maturity in large-scale linked-data deployments.

Full OWL 2 DL reasoning over the combined TBox and the ABox is not computationally feasible with Hermit on commodity hardware. We therefore use ELK [49] to validate the logical consistency of the knowledge graph. For query answering, we rely on a selected forward-rule expansion in Virtuoso, which physically expands the graph by applying subclass, subproperty, inverse, symmetric, equivalence, and transitivity rules; this expansion has not been demonstrated to reach a fixed point and is not complete OWL-Horst closure [58]. Two decades of work on scaling OWL reasoning—materialization-based approaches [58], query rewriting, modular reasoning, and consequence-based calculi such as ELK itself—provide several alternative points in the design space, and a more expressive reasoner could be applied to a sub-module of the TBox in future work.

6.2 Limitations and outstanding challenges

Several challenges remain open. First, the assay-aware control screen identifies 351 of 351,472 ASVs using 17 extraction-day blanks with batch links to Trip 5 profiles. Removing these candidates only from the 217 co-extracted Trip 5 biological profiles removes 2.19 % of reads in aggregate, and the unfiltered table remains canonical. Inference is bounded because the design provides only one blank per extraction batch, concentration data are absent, Trip 4 has no positive control and has an unsequenced blank affecting all or part of 23 sites, and Trip 5 has no 16S positive control. A reuser should use the control module and audit tables to select compatible controls by assay, stage and batch, and should not treat contamination as eliminated. Second, archived-soil pH is available for 712 admitted specimens under the specified CaCl_2 protocol, but coverage is incomplete and non-random and should not be interpreted as an in-situ field map. Third, the XRF sources lack machine-readable unit and non-detect semantics, and Trips 1–4 and Trip 5 use disclosed but different repeat-formula rules; field and laboratory XRF values cannot be assumed interchangeable, and their current comparison is site-level rather than same-aliquot. Fourth, more complex reasoning—for example over disjunctions arising from ENVO biome classifications or negative axioms about taxon absence—would require either a different reasoner or the extraction of a sub-module amenable to a more expressive calculus. Finally, the resource depends on upstream ontologies (SIO, ChEBI, ENVO, NCBI Taxonomy) that evolve

independently; keeping the deployed graph aligned with their releases is an ongoing maintenance task and a known reproducibility risk for any SPARQL query that bottoms out in a specific external IRI.

6.3 Sustainability and updates

The dataset is versioned and archived on Zenodo, and the deployment workflow is reproducible from the public repository. We intend to extend the resource as additional field expeditions are processed: new trips will be added as additional ABox modules without changes to the upper-level patterns. Upstream ontology updates will be tracked at scheduled intervals; re-materialization is scripted and runs as part of the deployment workflow, so an upstream release can be absorbed by re-running the workflow. Where an upstream change would alter the semantics of an existing query, we will record the affected competency questions in the changelog so that downstream users can audit the impact.

7 Data availability

Three project accessions appear in the project records, resolved against the live European Nucleotide Archive on 28 July 2026. The accessions are not yet reconciled, and none is currently a usable route to the data described here. **PRJEB106069** is the project against which all 1,268 submitted amplicon run rows are registered, but it returns no public record. **PRJEB104209** is a public umbrella study (“The genome of the Rub’ al Khali (Empty Quarter)”, secondary accession **ERP185514**) but carries no read runs of its own. **PRJNA1065643** is public and holds 85 amplicon read runs on Illumina MiSeq, an earlier partial deposition on a different instrument that is not the five-expedition release described here. Every run accession the project holds was checked individually, and none yet resolves to a public record; establishing the sample-level mapping requires **PRJEB106069** and its runs to be released. The staged release candidate contains the derived companion-analysis inputs (the paired PMA table, 150 CoverM profiles, eggNOG annotations for the 990-genome analytical subset and the measured-function input archive), which support their named downstream endpoints but do not reproduce raw-read processing, assembly, binning, dereplication, annotation or profile generation. Public deposition of the checksummed package and the underlying sequence resources remains an availability gate for submission.

8 Code availability

All source code for data processing, ontology engineering, RDF generation, validation, and the web portal is openly available at <https://github.com/bio-ontology-research-group/empty-quarter>. The repository includes Groovy scripts utilizing the OWL API 5 for OWL/RDF generation, Python scripts for data processing and statistical analysis, ShEx shape definitions for structural validation, SPARQL competency queries, React/FastAPI web application source code, Docker Compose configurations for deployment, and the `main.nf` Nextflow entry point with its pinned software environment.

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Author contributions

Add CRediT author contribution statements

Competing interests

The authors declare no competing interests.

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